A Survey on Image Priors for Image Denoising

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1 TASK ASSIGNMENT

- Yihui Wang is responsible for introduction, conclusion, future works part, and the combination of NSS and sparse representation method.
- Enming Guo is responsible for intended contributions, and DnCNN method.
- Leilong Fan is responsible for previous work, conclusion and Total Variation method.
- Tingyu Shi is responsible for motivation, problem statement, and Tikhonov method.

2 INTRODUCTION

2.1 Motivation

Digital images are vital to people's daily lives nowadays. Various applications depend on them, such as face recognition, object tracking, medical images, etc. However, for multiple reasons, image noise may exist when digital images are formed. According to [1], image noises mainly come from the following aspects:

- Different environmental conditions may impact image sensors during image acquisition.
- Low lighting conditions and elevated sensor temperatures can lead to noise.
- Noise may also stem from interference within the transmission channel.
- Additionally, dust particles on the scanner screen can introduce noise into the image.

Therefore, image denoising is an important step to preprocess images so that various applications mentioned can have better performance. This project focuses on image denoising performance based on different image priors.

2.2 Previous Works

In the domain of image denoising, significant progress has been made over the years through the development and application of various algorithms and models. These efforts aim to enhance image quality by reducing noise, thus improving the performance of subsequent image processing or analysis tasks. The field has evolved from simple spatial filtering techniques to sophisticated models that leverage mathematical and statistical frameworks.

2.2.1 Traditional Denoising Techniques. Initial approaches to image denoising were primarily focused on spatial and frequency domain filtering. Techniques such as Gaussian smoothing, median filtering, and Wiener filtering were widely used for their simplicity and effectiveness in reducing noise. However, these methods often struggle to preserve edges and fine details, leading to overly smoothed images.

2.2.2 Advanced Statistical Models. The advent of statistical models marked a significant advancement in image denoising. Methods like wavelet thresholding, which employs the wavelet transform to distinguish between noise and signal at different scales, have demonstrated improved denoising capabilities while better preserving image details. The work by Rohit Verma and Jahid Ali [1] provides a comparative study of various noise types and introduces efficient noise removal techniques, emphasizing the importance of tailored approaches to different denoise statistical models.

2.2.3 Non-Local Means and Self-Similarity. The Non-Local Means (NLM) algorithm introduced by Buades et al. [2] represents a paradigm shift by exploiting the self-similarity present within images. By averaging pixels based on the similarity of their neighborhoods rather than their spatial closeness, NLM can better preserve textures and structures. This concept of exploiting non-local information has been extended in various forms, including the BM3D algorithm, showcasing the potency of considering the global structure of images for denoising.

2.2.4 Sparse Representation and Dictionary Learning. Sparse representation has emerged as a powerful tool for image denoising, where images are represented as sparse linear combinations of elements from an overcomplete dictionary. The work of M. Aharon, M. Elad, and A. Bruckstein [3] on the K-SVD algorithm demonstrates how dictionaries can be learned from noisy images themselves to achieve impressive denoising results. This approach benefits from the adaptability to the unique structures and features within specific images.

2.2.5 Convex Optimization and Regularization Techniques. Convex optimization techniques can also be used for image denoising. Methods such as Tikhonov regularization and Total Variation (TV) denoising are based on formulating the denoising task as an optimization problem. These methods focus on balancing fidelity to the observed noisy image with the enforcement of smoothness or sparsity constraints.

2.3 Intended Contributions

Firstly, we formulate the image denoising problem as a convex optimization problem. Then, we also derived primal formulation, dual formulation, and KKT conditions.
Secondly, in the field of image denoising, choosing the appropriate image prior function(i.e. regularization term) is a key factor, as it represents the prior knowledge or assumptions about original images. In this project, we mainly focus on how different image prior functions can affect the performance of image denoising. Specifically, we present the following related to image prior functions:

- We introduce 4 image prior functions, which are Tikhonov Regularization(L_2 norm), Total Variation(L_1 norm), Non-local Self Similarity(NSS), Spare Representation.
- We introduce an image denoising algorithm proposed by Xu et al.[4], which combines by NSS and spare representation.
- We compare and contrast the denoising performance of different image prior functions.
- Summarize the characteristics usage scenarios of different image prior functions.

Thirdly, besides those classic methods mainly based on convex optimization, we also applied the convolutional neural network-based method DnCNN as a part of the comparative experiments. Unlike these conventional methods that rely on predefined priors and assumptions about the image characteristics, DnCNN leverages deep learning to predict the residual image $\hat{v}$. This approach allows DnCNN to implicitly extract the clean image using the operations within its hidden layers, without directly manipulating the noise characteristics. The inclusion of batch normalization further stabilizes and enhances the training performance of DnCNN, leading to potentially superior denoising results with improved generalization across different noise types and levels. The contribution of integrating DnCNN into the experiment lies in its ability to automatically learn and adapt to the underlying noise distribution, offering a more dynamic and effective solution for image denoising that could outperform traditional methods in both effectiveness and efficiency.

2.4 Organization of the Paper

The rest of this paper is organized as follows. In chapter 3, we begin by stating and describing the problem, including exposition of the primal formulation, the dual formulation, and the KKT conditions. In chapter 4, we describe our intended approaches, including L_1 and L_2 norm, the combination of NSS and Sparse priors, and DnCNN. In chapter 5, we then present our experimental results, and conclude with a discussion and future work.

3 STATEMENT OF THE PROBLEM

3.1 Primal Formulation

Let y represent the observed noisy image, x represent the original image without noise, n represent the noise. Then, the image denoising model can be formulated as the following:

$$y = x + n \tag{1}$$

In this project, we focus on additive Gaussian white noise(AGWN), therefore:

$$n \sim \mathcal{N}(0, \sigma) \tag{2}$$

Let $\hat{x}$ represent the image that we want to recover, then according to Maximum a Posterior(MAP):

$$\hat{x} = \underset{x}{argmax}\ P(x|y) \tag{3}$$

By using the Bayes theorem and the monotonicity of log function, we can reformulate equation(3) as the following:

$$\hat{x} = \underset{x}{argmin}\ \left(-log\left(P(y|x)\right) - log\left(P(x)\right)\right) \tag{4}$$

- $P(y|x)$ is the likelihood term. If x and y are similar, $P(y|x)$ will be close to 1, otherwise it will be close to 0. Therefore, $-log\left(P(y|x)\right)$ will be close to 0 if x and y are similar to each other. $-log\left(P(y|x)\right)$ will tend to be $+\infty$ if x and y are very different from each other. As a result, we use a fidelity term to represent $-log\left(P(y|x)\right)$, which is $\frac{1}{2}||x-y||_2^2$.
- $P(x)$ is the image prior. We usually use $R(x)$ to encode the term related to the image prior, which means that we will use $R(x)$ to represent $-log\left(P(x)\right)$.

We denote the height and the width of an image as H and W. Then $\Omega = \{(i,j)|i \in \mathbb{Z}, j \in \mathbb{Z}, 1 \le i \le H, 1 \le j \le W\}$ is a set of all the possible image coordinates. If x is an image, then $x_{i,j}((i,j) \in \Omega)$ represents the pixel value of image x at row i and column j. For an image, all the pixel values should be between 0 and 255.
From the above, the primal formulation is the following:

$$\begin{aligned} \hat{x} &= \underset{x}{argmin}\left(\frac{1}{2}||x-y||_2^2 + \Lambda R(x)\right) \\ s.t.&\quad x_{i,j} \in [0,255] \quad \forall (i,j) \in \Omega \end{aligned} \tag{5}$$

$\Lambda \ge 0$ is a scalar to balance the fidelity term and the regularization term.

3.2 Dual Formulation

First, we can reformulate primal formulation(5) mentioned above as the following:

$$\begin{aligned} \hat{x} &= \underset{x}{argmin}\left(\frac{1}{2}||x-y||_2^2 + \Lambda R(x)\right) \\ s.t.&\begin{cases} -x_{i,j} \le 0 & \forall (i,j) \in \Omega \\ x_{i,j} - 255 \le 0 & \forall (i,j) \in \Omega \end{cases} \end{aligned} \tag{6}$$

Then the Lagrangian function is the following:

$$\begin{aligned} L\left(x, \boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right) = \frac{1}{2}||x-y||_2^2 + \Lambda R(x) + \sum_{(i,j)\in\Omega} \lambda_{i,j}^{\mathrm{I}}\left(-x_{i,j}\right) \\ + \sum_{(i,j)\in\Omega} \lambda_{i,j}^{\mathrm{II}}\left(x_{i,j} - 255\right) \end{aligned} \tag{7}$$

Now, we need to differentiate $L\left(x, \boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right)$ with respect to x and set the derivative to 0.

$$\begin{aligned} \frac{\partial L\left(x, \boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right)}{\partial x} = (x-y) + \Lambda\frac{\partial R(x)}{\partial x} + \sum_{(i,j)\in\Omega} -\lambda_{i,j}^{\mathrm{I}} \\ + \sum_{(i,j)\in\Omega} \lambda_{i,j}^{\mathrm{II}} = 0 \end{aligned} \tag{8}$$

Since we use different image priors(i.e. $R(x)$), we do not have a closed form solution of x in equation(8). Let's assume that x^* is the solution of equation(8), which means the following equation is true:

$$(x^* - y) + \Lambda\frac{\partial R(x^*)}{\partial x} + \sum_{(i,j)\in\Omega} -\lambda_{i,j}^{\mathrm{I}} + \sum_{(i,j)\in\Omega} \lambda_{i,j}^{\mathrm{II}} = 0 \tag{9}$$

Then, the Lagrange dual function is the following:

$$g\left(\boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right) = \inf_{x} \; L\left(x, \boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right) = L\left(x^{*}, \boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right) \tag{10}$$

Then the dual problem is the following:

$$\begin{aligned} &\max_{\boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}} \; g\left(\boldsymbol{\lambda}^{\mathrm{I}}, \boldsymbol{\lambda}^{\mathrm{II}}\right) \\ &s.t. \begin{cases} \lambda^{\mathrm{I}}_{i,j} \geq 0 \quad \forall (i,j) \in \Omega \\ \lambda^{\mathrm{II}}_{i,j} \geq 0 \quad \forall (i,j) \in \Omega \end{cases} \end{aligned} \tag{11}$$

3.3 KKT conditions

3.3.1 *Primal Constraints.*

$$0 \leq x_{i,j} \leq 255 \quad \forall (i,j) \in \Omega \tag{12}$$

3.3.2 *Dual constraints.*

$$\begin{cases} \lambda^{\mathrm{I}}_{i,j} \geq 0 \quad \forall (i,j) \in \Omega \\ \lambda^{\mathrm{II}}_{i,j} \geq 0 \quad \forall (i,j) \in \Omega \end{cases} \tag{13}$$

3.3.3 *Complementary slackness.*

$$\begin{cases} \lambda^{\mathrm{I}}_{i,j} * (-x_{i,j}) = 0 \quad \forall (i,j) \in \Omega \\ \lambda^{\mathrm{II}}_{i,j} * (x_{i,j} - 255) = 0 \quad \forall (i,j) \in \Omega \end{cases} \tag{14}$$

3.3.4 *Gradient of Lagrangian with respect to x variables.*

$$\nabla_x \left(\frac{1}{2}||x-y||_2^2 + \Lambda R(x)\right) + \sum_{(i,j)\in\Omega} \lambda^{\mathrm{I}}_{i,j} \nabla_x \left(-x_{i,j}\right) + \sum_{(i,j)\in\Omega} \lambda^{\mathrm{II}}_{i,j} \nabla_x \left(x_{i,j} - 255\right) = 0 \tag{15}$$

$$(x-y) + \Lambda\left(\nabla_x R(x)\right) + \sum_{(i,j)\in\Omega} -\lambda^{\mathrm{I}}_{i,j} + \sum_{(i,j)\in\Omega} \lambda^{\mathrm{II}}_{i,j} = 0 \tag{16}$$

4 INTENDED APPROACHES

4.1 Tikhonov Regularization

Tikhonov Regularization was first introduced by Andrey Tikhonov in a book called "Solutions of ill-posed problems"[5]. This technique was introduced to solve ill-posed ordinary least square linear regression.

$$\begin{aligned} &Formulation1 : \min_{x} ||Ax-b||_2^2 \\ &Formulation2 : \min_{x} ||Ax-b||_2^2 + ||\Gamma x||_2^2 \end{aligned}$$

It is possible that formulation 1 above does not have a solution or has infinite number of solutions. To solve this problem, a regularization term was introduced (i.e. $||\Gamma x||_2^2$) to give preference to a certain solution. For image denoising, the typical choice of Tikhonov regularization term(i.e. image prior) is the following:

$$R(x) = \frac{1}{2}||\nabla x||_2^2 \tag{17}$$

Here, we put ∇x inside of L_2 norm instead of x because this can better to preserve edges and details. Therefore, the minimization problem is the following:

$$\hat{x} = \underset{x}{argmin}\left(\frac{1}{2}||x-y||_2^2 + \frac{\Lambda}{2}||\nabla x||_2^2\right) \tag{18}$$

Now, let $f(x)$ be the following:

$$f(x) = \frac{1}{2}||x-y||_2^2 + \frac{\Lambda}{2}||\nabla x||_2^2 \tag{19}$$

Now, we set the derivative of $f(x)$ with respect to x be 0 and then solve x.

$$\frac{\partial f(x)}{x} = (x-y) + \Lambda \Delta x = 0 \tag{20}$$

In equation(20), Δx represents the laplacian operator of the image. It is defined as the following:

$$\Delta x_{i,j} = 4x_{i,j} - x_{i-1,j} - x_{i+1,j} - x_{i,j-1} - x_{i,j+1} \tag{21}$$

Now, we can substitute equation(21) into equation(20). Then, we have the following:

$$x_{i,j} + 4\Lambda x_{i,j} = y_{i,j} + \Lambda(x_{i-1,j} + x_{i+1,j} + x_{i,j-1} + x_{i,j+1}) \tag{22}$$

$$x_{i,j} = \frac{y_{i,j} + \Lambda(x_{i-1,j} + x_{i+1,j} + x_{i,j-1} + x_{i,j+1})}{1+4\Lambda} \tag{23}$$

By using a Jacobi iterative method mentioned in this paper[6], we can solve $x_{i,j}$ as the following:

$$x^{k+1}_{i,j} = \frac{y_{i,j} + \Lambda(x^{k}_{i-1,j} + x^{k}_{i+1,j} + x^{k}_{i,j-1} + x^{k}_{i,j+1})}{1+4\Lambda} \quad k = 1, 2, 3 \cdots \tag{24}$$

For boarder pixels, we used mirror padding so that they can be calculated as mentioned in equation(24).
Figure 1 shows how Λ values can affect the image denoising performance. Larger Λ value can better remove the noise, but also oversmooth image edges at the same.

4.2 Total Variation Denoising

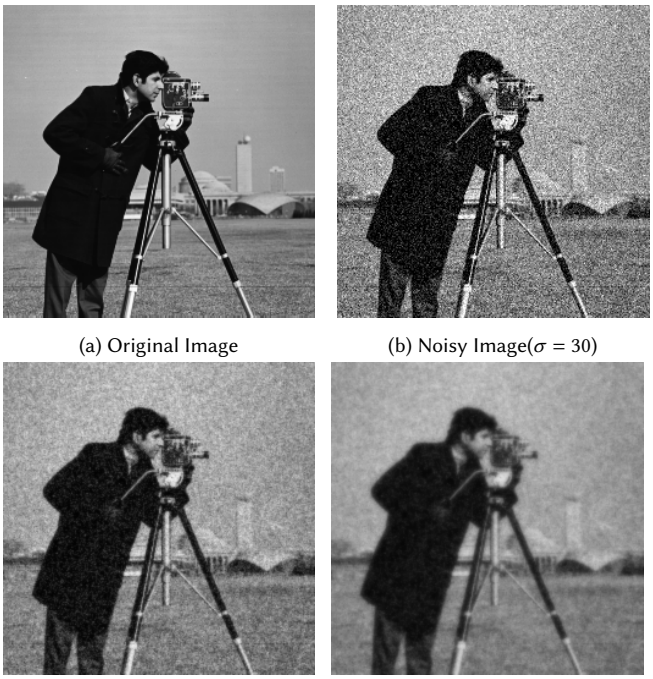
For Total Variation Denoising, the regularization term(image prior) is the L_1 norm of the image derivative.

$$R(x) = ||\nabla x||_1 \tag{25}$$

Then, the minimization problem is the following:

$$\hat{x} = \underset{x}{argmin}\left(\frac{1}{2}||x-y||_2^2 + \frac{\Lambda}{2}||\nabla x||_1\right) \tag{26}$$

As mentioned in this paper[7], many algorithms have been proposed to solve Total Variation minimization problem, such algorithms based on duality[8], Newton-based methods[9], graph cuts[10], etc. To solve this minimization problem, we plan to employ Chambolle's projection algorithm[11], an efficient method for minimizing the total variation (TV) of an image to achieve denoising. This algorithm is based on the dual formulation of TV minimization and is particularly effective for denoising images affected by additive white Gaussian noise (AWGN).
Chambolle's algorithm transforms the original minimization problem into a more solvable form by converting it into its dual problem.



(a) Original Image (b) Noisy Image($\sigma = 30$)

(c) Denoised Image($\Lambda = 0.5$) (d) Denoised Image($\Lambda = 2$)

Fig. 1. Tikhonov Regularization with Different Λ values

The core idea of the algorithm is to iteratively update the dual variable p, and approximate the optimal solution through the following iterative scheme:

$$p^{(k+1)} = \frac{p^{(k)} + \tau\nabla(\mathrm{div}p^{(k)} - \Lambda y)}{1 + \tau|\nabla(\mathrm{div}p^{(k)} - \Lambda y)|}, \tag{27}$$

where $\tau > 0$ is the time step parameter, and div represents the divergence operator. The algorithm iterates until convergence, that is, when the maximum change between $p^{(k+1)}$ and $p^{(k)}$ is less than a predetermined tolerance threshold.

We also apply an adaptive method for selecting the regularization parameter Λ, to balance the preservation of image details and noise removal. By this method, we expect to effectively remove noise in the image while maintaining its structure and edges.

4.3 The combination of NSS and Sparse Image prior

Fan et al.[12] offer a thorough overview of various image denoising techniques, including the NCSR model that effectively combines Non-Local Self-Similarity (NSS) and Sparse Representation for enhanced performance.

4.3.1 Non-Local Self-Similarity (NSS). NSS is a concept used in image processing that refers to the observation that segments within a image can exhibit similar patterns, regardless of their location within the image[13]. It exploits the repetitive nature of textures and structures across different parts of one image. A well-known application of NSS is Non-Local Means (NLM)[2] algorithm, where each pixel's value is updated based on a weighted average of all other pixels in the image, with weights determined by the similarity between patches centered around the pixels.

Let $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ be image patches centered at x_i and x_j, respectively. Let $w_{i,j}$ be the weight from $\boldsymbol{x}_j$ to $\boldsymbol{x}_i$, which is computed as follows:

$$w_{i,j} = \frac{1}{c_i}\exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2}{h}\right) \tag{28}$$

where c_i denotes a normalization factor, and h represents a filtering parameter. Then, the NSS prior can be defined as:

$$R_{\mathrm{NSS}}(\mathbf{x}) = \sum_{\mathbf{x}_i\in\mathbf{x}} \|\mathbf{x}_i - \mathrm{NLM}(\mathbf{x}_i)\|_2^2 = \sum_{\mathbf{x}_i\in\mathbf{x}} \left\|\mathbf{x}_i - \mathbf{w}_i^T\mathbf{K}_i\right\|_2^2 \tag{29}$$

where κ_i and $\mathbf{w}_i$ denote column vectors; the former contains the central pixels around $\mathbf{x}_i$, and the latter contains all the corresponding weights $w_{i,j}$. Currently, the focus of research in image denoising has transitioned from local to non-local methods.

4.3.2 Sparse Representation. Sparse representation present an image over an over-complete dictionary D with L1-norm regularization on the vector, indicating the following model:

$$\hat{\alpha} = \underset{\alpha}{argmin}\ \|y - D\alpha\|_2^2 + \Lambda\|\alpha\|_1 \tag{30}$$

where $\hat{\alpha}$ is a matrix which contains vectors of sparse coefficient.

As a dictionary learning approach, the sparse representation model can be adapted based on a given dataset or directly from the noisy image in question. This adaptation is achieved through the K-singular value decomposition (K-SVD) algorithm. K-SVD denoising operates on the principle of concurrently learning the dictionary D from a noisy image y and involves addressing a joint optimization problem that seeks to find the dictionary that not only sparsely represents the image but also effectively reduces the noise:

$$\underset{x,D,\alpha}{argmin}\ \lambda\|y - x\|_2^2 + \sum_i \|R_i x - D\alpha_i\|_2^2 + \sum_i \mu_i\|\alpha_i\|_1 \tag{31}$$

where R_i represents a matrix that extracts the patch $\boldsymbol{x}_i$ from the image x at a specific location i.

Sparse representation models with learned dictionaries perform better than designed dictionaries, because it is able to adapt to the specific structures present with an image allowing for a more accurate representation. However, in the case of high noise, sparse representation with local information can be significantly corrupted, which means the result is not that efficient.

When introducing NSS prior, the sparsity from self-similarity properties is wildely used for image denoising. The NCSR model is one of the representative work:

$$\alpha_y = \arg\min_{\alpha} \|y - D\alpha\|_2^2 + \Lambda\sum_{i=1}^{N}\|\alpha_i - \beta_i\|_1 \tag{32}$$

where β_i is a good estimation of α. Then, for each x_i, β_i:

$$\beta_i = \sum_{q\in S_i} w_{i,q}\alpha_{i,q} \tag{33}$$

The NCSR model successfully incorporates the NSS into the sparse representation work, making it frequently adopted in image denoising. This integration allows the model to exploit the repetitive nature of textures and patterns within the image, thus enhancing the denoising process.

4.3.3 Patch Group NSS Prior. As we introduced before, image NSS has been widely used in patched based image denoising. Here, we present a patch based prior method for image denoising proposed by Xu et al. [4]. Unlike other methods that only the NSS information of the noisy image is exploited, this method proposes a Patch Group(PG) based NSS prior learning scheme, which is Patch-Group Gaussian Mixture Model(PG-GMM), to learn explicit NSS models from natural images for high performance denoising. This method also includes sparse representation.

This method first performs Patch Group and Group Mean Subtraction. This process involves selecting a local patch from a clean image and finding the top M most similar non-local patches throughout the image from a PG. By doing this, the mean vectors can be represented:

$$\mathbf{X_n} \triangleq \{x_{n,m} - \frac{1}{M}\sum_{m=1}^{M} x_m\}, n = 1, \ldots, N \tag{34}$$

$\mathbf{X_n}$ involves many NSS information of natural images, then the problem turns out to be how to learn prior model from it. Armed with PG-GMM, the goal is to learn $\{\mathcal{N}(\mu_k, \Sigma_k)\}$ from PGs. The likelihood of $\mathbf{X_n}$ is:

$$P(\mathbf{X}_n) = \sum_{k=1}^{K} \pi_k \prod_{m=1}^{M} \mathcal{N}(\mathbf{x}_{n,m}|\mu_k, \Sigma_k). \tag{35}$$

The overall objective likelihood is $\mathcal{L} = \prod_{n=1}^{N} P(\mathbf{X}_n)$. Representing this equation with log:

$$\ln \mathcal{L} = \sum_{n=1}^{N} \ln\left(\sum_{k=1}^{K} \pi_k \prod_{m=1}^{M} \mathcal{N}(\hat{\mathbf{x}}_{n,m}|\mu_k, \Sigma_k)\right). \tag{36}$$

Then, it is feasible to use EM algorithm to optimize the above equation. In the E-Step:

$$\gamma_{nk} = \frac{\pi_k \prod_{m=1}^{M} \mathcal{N}(\mathbf{x}_{n,m}|\mu_k, \Sigma_k)}{\sum_{l=1}^{K} \pi_l \prod_{m=1}^{M} \mathcal{N}(\mathbf{x}_{n,m}|\mu_l, \Sigma_l)} \tag{37}$$

While in the M step, for each PG, $\sum_{m=1}^{M} x_{n,m} = \mathbf{0}$:

$$\mu_k = \frac{\sum_{n=1}^{N} \gamma_{nk} \sum_{m=1}^{M} x_{n,m} x_{n,m}^T}{\sum_{n=1}^{N} \gamma_{nk}} = \mathbf{0}, \tag{38}$$

$$\Sigma_k = \frac{\sum_{n=1}^{N} \gamma_{nk} \sum_{m=1}^{M} x_{n,m} x_{n,m}^T}{\sum_{n=1}^{N} \gamma_{nk}}. \tag{39}$$

By iterating between the E-step and M-step, the model parameters can be updated in each iteration, where the value of log-likelihood is guaranteed to increase. The EM algorithm should converge.

In the denoising stage, for each local patch in a noisy image, similar patches are searched within a window to form a PG. The group mean of these patches is calculated and subtracted from each patch to obtain the mean subtracted PG, which is then used to recover the clean PG using learned NSS prior models. The selection can be presented:

$$P(k|\mathbf{Y}) = \frac{\prod_{m=1}^{M} \mathcal{N}(\mathbf{y}_m|\mathbf{0}, \Sigma_k + \sigma^2\mathbf{I})}{\sum_{l=1}^{K} \prod_{m=1}^{M} \mathcal{N}(\mathbf{y}_m|\mathbf{0}, \Sigma_l + \sigma^2\mathbf{I})} \tag{40}$$

where $\mathcal{N}(\mathbf{y}_m|\mathbf{0}, \Sigma_k + \sigma^2\mathbf{I})$ represents the likelihood that the m-th patch y_m of the noisy image is generated by a Gaussian distribution with mean zero and covariance $\Sigma_k + \sigma^2\mathbf{I}$. The covariance is adjusted by adding $\sigma^2\mathbf{I}$ to account for the noise variance σ^2, where $\mathbf{I}$ is the identity matrix.

Substituting the log-likelihood, we have:

$$\ln P(k|\mathbf{Y}) = \sum_{m=1}^{M} \ln \mathcal{N}(\mathbf{y}_m|\mathbf{0}, \Sigma_k + \sigma^2\mathbf{I}) - \ln C \tag{41}$$

For each patch y_m, like what we illustrate in sparse representation, the denoising model is:

$$\min_{\alpha} \|\mathbf{y}_m - D\alpha\|_2^2 + \Lambda\|\alpha\|_1. \tag{42}$$

By inferring arguments, the optimal solution can be:

$$\hat{\alpha} = \underset{\alpha}{argmin}\|\mathbf{y}_m - D\alpha\|_2^2 + \sum_{i=1}^{p^2} \frac{c * \sqrt{2\sigma^2}}{\lambda_i}|\alpha_i|. \tag{43}$$

With $\hat{\alpha}$ solved in equation(43), the clean patch can be estimated as

$$\hat{x}_m = D\hat{\alpha} + \mu_y \tag{44}$$

Then the final clean image can be formed by aggregating all the estimated patches mentioned in equation(44)

4.4 DnCNN

Zhang et al. [14] refers to a denoising convolutional neural network called DnCNN, which is designed to predict the residual image $\hat{v}$ rather than directly outputing the denoised image $\hat{x}$, where $\hat{v}$ is the difference between the noisy observation and the latent clean image. The DnCNN model implicitly removes the latent clean image with the operations in the hidden layers. The batch normalization technique is further introduced to stabilize and enhance the training performance of DnCNN.

4.4.1 Network Architecture Design. Regarding the network architecture, DnCNN adapt the VGG network [15] by modifying it to be appropriate for image denoising tasks. The depth of the modified network is determined by considering the effective patch sizes employed in state-of-the-art denoising methods.

A. *Network Depth.* Adhering to the principles outlined in [15], the size of the convolutional filters is set to be 3×3 while all pooling layers are removed. Therefore, the receptive field of DnCNN with depth d is $(2d + 1) \times (2d + 1)$. Increasing the receptive field size allows the utilization of context information from a larger image region. It has been pointed out that the receptive field size of denoising neural networks correlates with the effective patch size of denoising methods [16, 17]. Moreover, higher noise levels usually require larger effective patch sizes to capture more contextual information for effective restoration [18]. Thus, by fixing the noise level

$\sigma = 25$, the effective patch sizes of several leading denoising methods are analyzed to guide the depth design of the DnCNN architecture.

B. *Network Architecture.* The input to the DnCNN is a noisy observation $y = x + v$, where x represents the latent clean image and v is the noise component. Discriminative denoising models such as MLP [19] and CSF [20] aim to learn a mapping function $\mathcal{F}(y) = x$ to directly predict the latent clean image. However, for the DnCNN, a residual learning formulation is adopted to train a residual mapping $\mathcal{R}(y) \approx v$, from which the clean image can be obtained by $x = y - \mathcal{R}(y)$. Formally, the averaged mean squared error between the desired residual images and the estimated residuals from the noisy input:

$$\ell(\Theta) = \frac{1}{2N} \sum_{i=1}^{N} |\mathcal{R}(y_i; \Theta) - (y_i - x_i)|_F^2 \tag{45}$$

is employed as the loss function to learn the trainable parameters Θ of the DnCNN. Here, $(y_i, x_i)_{i=1}^N$ represents N noisy-clean training image (patch) pairs. Fig. 2 illustrates the architecture of the proposed DnCNN for learning the residual mapping $\mathcal{R}(y)$. In the following sections, the architecture of the DnCNN and the strategy for reducing boundary artifacts are explained.

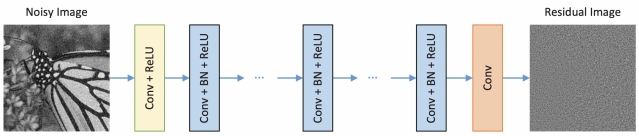


Fig. 2. The architecture of the proposed DnCNN network

4.4.2 Model Learning. During the model training process, we employ a residual learning formulation and integrate it with batch normalization techniques, enabling faster training convergence and enhanced denoising performance.

The network architecture illustrated in Fig. 2 can be employed to train either the original mapping $\mathcal{F}(y)$ to directly predict the clean image x, or the residual mapping $\mathcal{R}(y)$ to estimate the noise component v. According to the analysis in [21], when the original mapping is close to an identity mapping, optimizing the residual mapping tends to be much easier. It is important to note that the noisy observation y is generally more similar to the latent clean image x than to the residual noise image v, especially in cases where the noise level is relatively low. Consequently, $\mathcal{F}(y)$ would be closer to an identity mapping compared to $\mathcal{R}(y)$, making the residual learning formulation more suitable for the task of image denoising.

This work employs a residual learning formulation and integrates it with batch normalization for image denoising. Experiments using gradient-based optimization algorithms like stochastic gradient descent (SGD) and Adam demonstrate that the combination of residual learning and batch normalization leads to faster convergence and better denoising performance compared to conventional approaches. It turns out that residual learning and batch normalization can benefit from each other, and their integration is effective in speeding up the training and boosting the denoising performance. Besides, the optimization algorithms themselves (SGD or Adam) are less crucial than the integration of residual learning and batch normalization in achieving state-of-the-art denoising results.

4.5 Algorithm Time Complexity

- Tikhonov Regularization: Let the image size be $M \times N$, then the time complexity is $O(MNT)$, where T is the number of iterations.
- Total Variation: Chambolle's Projection Algorithm for total variation denoising has a per-iteration complexity of $O(N)$, where N is the number of pixels in the image. This complexity is due to operations such as gradient computation, divergence computation, and projection, each requiring $O(N)$ operations. The total complexity of the algorithm is $O(IN)$, with I being the number of iterations until convergence. The number of iterations, I, depends on the convergence criteria and the image characteristics, making the total complexity variable. However, I is typically not very large, rendering the algorithm efficient for image denoising tasks.
- PGPD: In the training stage of PG-GMM, for N PGs and each of them has M patches, the total number of patches is $M \times N$. In the M-step, the cost is $O(p^4MN)$, since the mean of Gaussian component is 0. In the E-step the cost is $O(p^6MN)$. Assume we have T iterations in total, the overall computational complexity is $O(p^6MNT)$.
  In the denoising stage, the cost for Gaussian component selection is $O(p^6NMK)$. The cost for closed-form weighted sparse coding is $O(p^4NMK)$. Assume we have T iterations, the overall computational complexity is $O(p^6NMKT)$.
- DnCNN: For the denoising task of 256x256 single-channel images, the computational complexity of the DnCNN model can be estimated by the number of multiply-add operations. At a noise level of 25, the receptive field size is set to 35x35 in the paper, corresponding to a network depth of 17 layers. Considering factors such as input size, kernel size, number of feature map channels, and the number of intermediate layers, the total computational complexity of the model for a single 256x256 grayscale image is approximately 6.86×10^{11} multiply-add operations. Although this complexity is relatively high, it is manageable for modern GPUs.

4.6 Evaluation Metrics

To evaluate the image denoising performance, we need to compare the difference between the original image and the noisy image with the difference between the original image and denoised image. To calculate the difference between two images, we use the following two metrics:

- Peak Signal-to-Noise Ratio(PSNR): Let x and y represent two images, then MSE between two images are defined as the following:

$$MSE(x, y) = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} \left(x_{i,j} - y_{i,j}\right)^2 \tag{46}$$

Knowing the MSE between 2 images, PSNR is defined as the following:

$$PSNR(x, y) = 20 \cdot log_{10}\left(\frac{255}{MSE(x, y)}\right) \tag{47}$$

The higher PSNR value, the more similar the two images are.

- Structural Similarity Index(SSIM) was first introduced in this paper[22]. Unlike traditional error metrics like PSNR, which rely solely on pixel-by-pixel comparisons, SSIM takes into account the structural information and visual perception aspects of the images. As a result, this metric can better simulate the human vision system. The equation of SSIM is the following:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_1)} \tag{48}$$

where μ_x and μ_y represent the mean of image x and y, σ_x and σ_y represent the variance of image x and y, and σ_{xy} represents the covariance between image x and y.

5 RESULTS

We did experiments on the following four denoising techniques:

- Tikhonov Regularization(L_2 norm prior)
- Total Variation(L_1 norm prior)
- Patch Group Based NSS Prior(NSS + Sparse Priors)
- DnCNN

We put our experiments in Appendix. The details are the following:

- Figure 3 shows 5 original images that we used for our experiments.
- Figure 4 shows all the noisy images. In our experiment, we add Guassian noise with 5 different standard deviation values, which are 10, 20, 30, 40, and 50.
- Figure 5 shows denoised images with Tikhonov Regularization.
- Figure 6 shows denoised images with Total Variation.
- Figure 7 shows denoised images with Patch Group Based NSS Prior.
- Figure 8 shows denoised images with DnCNN.
- Table 1 shows the denoising performance of image "Cameraman".
- Table 2 shows the denoising performance of image "House".
- Table 3 shows the denoising performance of image "Bird".
- Table 4 shows the denoising performance of image "Fighter Jet".
- Table 5 shows the denoising performance of image "Vegetable".

5.1 Conclusion

Our survey covers a range of methods including Tikhonov regularization, Total Variation Denoising, the fusion of Non-Local Self-Similarity (NSS) with sparse representation, and deep learning models, particularly DnCNN. Through extensive experiments on various natural images at different noise levels, we have rigorously evaluated the performance of each method based on SSIM and PSNR metrics.

Our findings indicate that, although traditional methods like Tikhonov regularization and total variation denoising can significantly improve image quality, they are sometimes surpassed by more complex models utilizing NSS and sparse representation. These models excel in preserving intricate details and textures of images, demonstrating the importance of adopting prior knowledge about image structure. Moreover, the DnCNN model, with its deep learning capabilities, has shown exceptional efficiency in denoising. It effectively learns the underlying noise distribution and achieves outstanding results, as evidenced by its consistently high SSIM and PSNR scores across different noise levels and images.

By visually checking the denoised images, we have the following findings:

- After denoising with Tikhonov Regularization, we can still observe lots of noise in the image, especially when σ is high.
- After denoising, all four algorithms can preserve the edges in images.
- In terms of preserving details of images, DnCNN has the best performance.

The comparative analysis underscores the critical role of selecting the appropriate denoising technique based on the specific requirements of the application and the characteristics of the noise present. Each method demonstrates unique strengths and limitations, highlighting the necessity of adopting a nuanced approach to image denoising.

5.2 Possible Future Works

The following are possible future works

- Firstly, we plan to address the denoising of color images, which introduces complex challenges not present in grayscale image processing. Color images require the preservation of color fidelity and the handling of noise across different color channels. Developing denoising algorithms that can effectively manage these aspects while maintaining image quality is of paramount importance.
- Secondly, since Posisson noise is also a type of additive noise, we can also do experiment to evaluate the denoising performance on Possion noise.
- Lastly, in addition to the successful combination of NSS and sparse representation techniques, we plan to explore new combinations of image prior functions to further advance denoising capabilities. Examples include combining Tikhonov Regularization with Total Variation, integrating deep learning features with sparse representation, etc.

By pursuing these directions, we hope to enhance the practical applicability and effectiveness of denoising algorithms across a wider range of imaging conditions and applications.

REFERENCES

[1] Rohit Verma and Jahid Ali. A comparative study of various types of image noise and efficient noise removal techniques. *International Journal of advanced research in computer science and software engineering*, 3(10), 2013.

[2] Antoni Buades, Bartomeu Coll, and J-M Morel. A non-local algorithm for image denoising. In *2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05)*, volume 2, pages 60–65. Ieee, 2005.

[3] Michal Aharon, Michael Elad, and Alfred Bruckstein. K-svd: An algorithm for designing overcomplete dictionaries for sparse representation. *IEEE Transactions on signal processing*, 54(11):4311–4322, 2006.

[4] Jun Xu, Lei Zhang, Wangmeng Zuo, David Zhang, and Xiangchu Feng. Patch group based nonlocal self-similarity prior learning for image denoising. In *Proceedings of the IEEE international conference on computer vision*, pages 244–252, 2015.

[5] Andrey N. Tikhonov and Vasiliy Y. Arsenin. Solutions of ill-posed problems. *(No Title)*, 1977.

[6] Beilei Tong and Qianshun Chang. Research on three common regularized image processing models. 2018.

[7] Joan Duran, Bartomeu Coll, and Catalina Sbert. Chambolle's projection algorithm for total variation denoising. *Image processing on Line*, 2013:311–331, 2013.

[8] Mingqiang Zhu, Stephen J Wright, and Tony F Chan. Duality-based algorithms for total-variation-regularized image restoration. *Computational Optimization and Applications*, 47(3):377–400, 2010.

[9] Michael K Ng, Liqun Qi, Yu-fei Yang, and Yu-Mei Huang. On semismooth newton's methods for total variation minimization. *Journal of Mathematical Imaging and Vision*, 27(3):265–276, 2007.

[10] Antonin Chambolle and Jérôme Darbon. On total variation minimization and surface evolution using parametric maximum flows. *International journal of computer vision*, 84(3):288–307, 2009.

[11] Antonin Chambolle. An algorithm for total variation minimization and applications. *Journal of Mathematical imaging and vision*, 20:89–97, 2004.

[12] Linwei Fan, Fan Zhang, Hui Fan, and Caiming Zhang. Brief review of image denoising techniques. *Visual Computing for Industry, Biomedicine, and Art*, 2:1–12, 2019.

[13] Guy Gilboa and Stanley Osher. Nonlocal operators with applications to image processing. *Multiscale Modeling & Simulation*, 7(3):1005–1028, 2009.

[14] Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising. *IEEE transactions on image processing*, 26(7):3142–3155, 2017.

[15] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition, 2015.

[16] Viren Jain and Sebastian Seung. Natural image denoising with convolutional networks. In D. Koller, D. Schuurmans, Y. Bengio, and L. Bottou, editors, *Advances in Neural Information Processing Systems*, volume 21. Curran Associates, Inc., 2008.

[17] Harold C Burger, Christian J Schuler, and Stefan Harmeling. Image denoising: Can plain neural networks compete with bm3d? In *2012 IEEE conference on computer vision and pattern recognition*, pages 2392–2399. IEEE, 2012.

[18] Anat Levin and Boaz Nadler. Natural image denoising: Optimality and inherent bounds. In *CVPR 2011*, pages 2833–2840, 2011.

[19] Harold C. Burger, Christian J. Schuler, and Stefan Harmeling. Image denoising: Can plain neural networks compete with bm3d? In *2012 IEEE Conference on Computer Vision and Pattern Recognition*, pages 2392–2399, 2012.

[20] Uwe Schmidt and Stefan Roth. Shrinkage fields for effective image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2774–2781, 2014.

[21] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition, 2015.

[22] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004.

6 APPENDIX

σ	Noisy image		Tikhonov		Total Variation		NSS-Sparse		DnCNN	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
10	0.64	28.33	0.73	30.05	0.91	32.67	0.93	34.09	0.92	33.51
20	0.41	22.44	0.63	26.26	0.84	29.11	0.86	30.32	0.88	30.68
30	0.30	19.05	0.56	24.41	0.81	27.30	0.82	28.48	0.84	29.00
40	0.24	16.66	0.51	23.13	0.78	26.05	0.80	27.27	0.82	27.87
50	0.19	14.88	0.48	22.24	0.76	25.14	0.77	26.39	0.79	27.01
Avg.	0.36	20.25	0.58	25.22	0.82	28.05	0.84	29.31	0.85	29.61

Table 1. Cameraman Image Denoising Result

σ	Noisy image		Tikhonov		Total Variation		NSS-Sparse		DnCNN	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
10	0.60	28.13	0.82	32.21	0.93	34.70	0.91	36.47	0.90	36.19
20	0.35	22.13	0.71	29.09	0.85	31.88	0.87	33.79	0.87	33.72
30	0.23	18.69	0.66	27.35	0.83	30.22	0.84	32.17	0.85	32.06
40	0.17	16.30	0.62	26.25	0.80	29.10	0.83	30.96	0.83	30.74
50	0.13	14.47	0.59	25.23	0.79	28.24	0.81	29.88	0.80	29.61
Avg.	0.30	19.94	0.68	28.02	0.84	30.83	0.85	32.65	0.85	32.46

Table 2. House Image Denoising Result

σ	Noisy image		Tikhonov		Total Variation		NSS-Sparse		DnCNN	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
10	0.68	28.25	0.77	29.99	0.92	32.46	0.92	33.48	0.92	33.14
20	0.46	22.48	0.67	26.21	0.86	28.86	0.86	29.93	0.87	30.16
30	0.34	19.09	0.59	24.24	0.82	27.07	0.82	28.05	0.84	28.45
40	0.27	16.76	0.54	22.83	0.79	25.87	0.80	26.81	0.81	27.26
50	0.23	15.04	0.50	21.80	0.76	24.81	0.78	25.91	0.78	26.33
Avg.	0.40	20.32	0.61	25.01	0.83	27.81	0.84	28.84	0.84	29.07

Table 3. Bird Image Denoising Result

σ	Noisy image		Tikhonov		Total Variation		NSS-Sparse		DnCNN	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
10	0.67	28.12	0.75	29.95	0.92	32.42	0.92	33.41	0.93	33.31
20	0.44	22.22	0.66	26.34	0.87	28.69	0.87	29.74	0.88	29.98
30	0.33	18.86	0.60	24.62	0.82	26.64	0.84	27.67	0.85	28.07
40	0.26	16.73	0.56	23.50	0.79	25.29	0.80	26.32	0.82	26.73
50	0.21	15.11	0.53	22.59	0.76	24.34	0.77	25.33	0.79	25.74
Avg.	0.38	20.21	0.62	25.40	0.83	27.48	0.84	28.49	0.85	28.77

Table 4. Fighter Jet Image Denoising Result

σ	Noisy image		Tikhonov		Total Variation		NSS-Sparse		DnCNN	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
10	0.68	28.26	0.82	30.55	0.92	33.23	0.92	34.61	0.93	34.52
20	0.46	22.48	0.71	27.01	0.87	29.85	0.88	31.33	0.89	31.47
30	0.34	19.09	0.65	25.21	0.84	28.01	0.85	29.32	0.85	29.58
40	0.27	16.76	0.62	24.02	0.81	26.70	0.82	27.91	0.82	28.21
50	0.23	15.04	0.58	23.16	0.79	25.67	0.79	26.76	0.79	27.13
Avg.	0.40	20.33	0.68	25.99	0.85	28.69	0.85	29.99	0.86	30.18

Table 5. Vegetable Image Denoising Result



(a) Cameraman (b) House (c) Bird (d) Fighter Jet (e) Vegetable

Fig. 3. Original Images



(a) σ = 10 (b) σ = 20 (c) σ = 30 (d) σ = 40 (e) σ = 50

(f) σ = 10 (g) σ = 20 (h) σ = 30 (i) σ = 40 (j) σ = 50

(k) σ = 10 (l) σ = 20 (m) σ = 30 (n) σ = 40 (o) σ = 50

(p) σ = 10 (q) σ = 20 (r) σ = 30 (s) σ = 40 (t) σ = 50

(u) σ = 10 (v) σ = 20 (w) σ = 30 (x) σ = 40 (y) σ = 50

Fig. 4. Noisy Images



(a) σ = 10 (b) σ = 20 (c) σ = 30 (d) σ = 40 (e) σ = 50

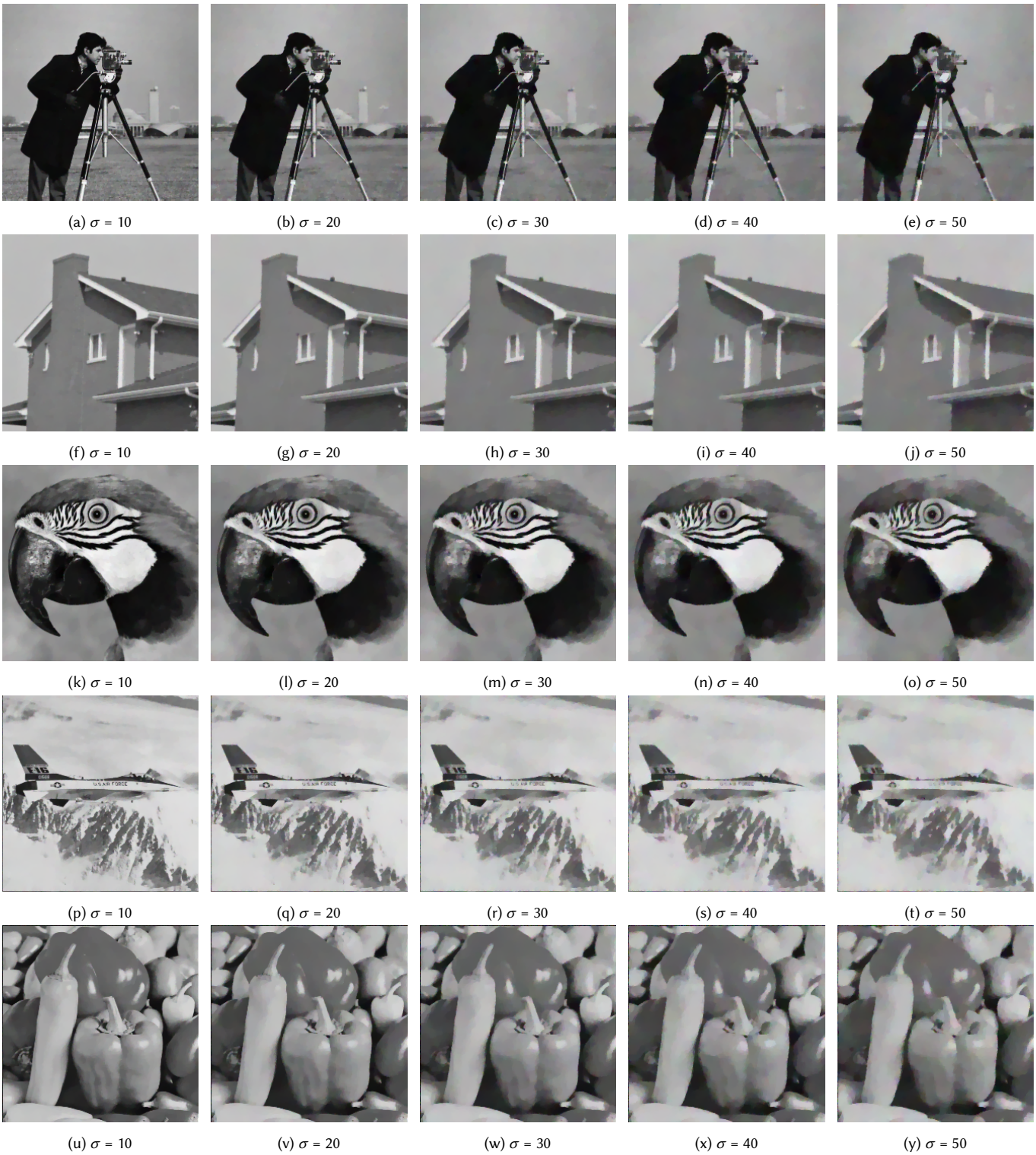
(f) σ = 10 (g) σ = 20 (h) σ = 30 (i) σ = 40 (j) σ = 50

(k) σ = 10 (l) σ = 20 (m) σ = 30 (n) σ = 40 (o) σ = 50

(p) σ = 10 (q) σ = 20 (r) σ = 30 (s) σ = 40 (t) σ = 50

(u) σ = 10 (v) σ = 20 (w) σ = 30 (x) σ = 40 (y) σ = 50

Fig. 5. Tikhonov Regularization Denoised Images



(a) σ = 10 (b) σ = 20 (c) σ = 30 (d) σ = 40 (e) σ = 50

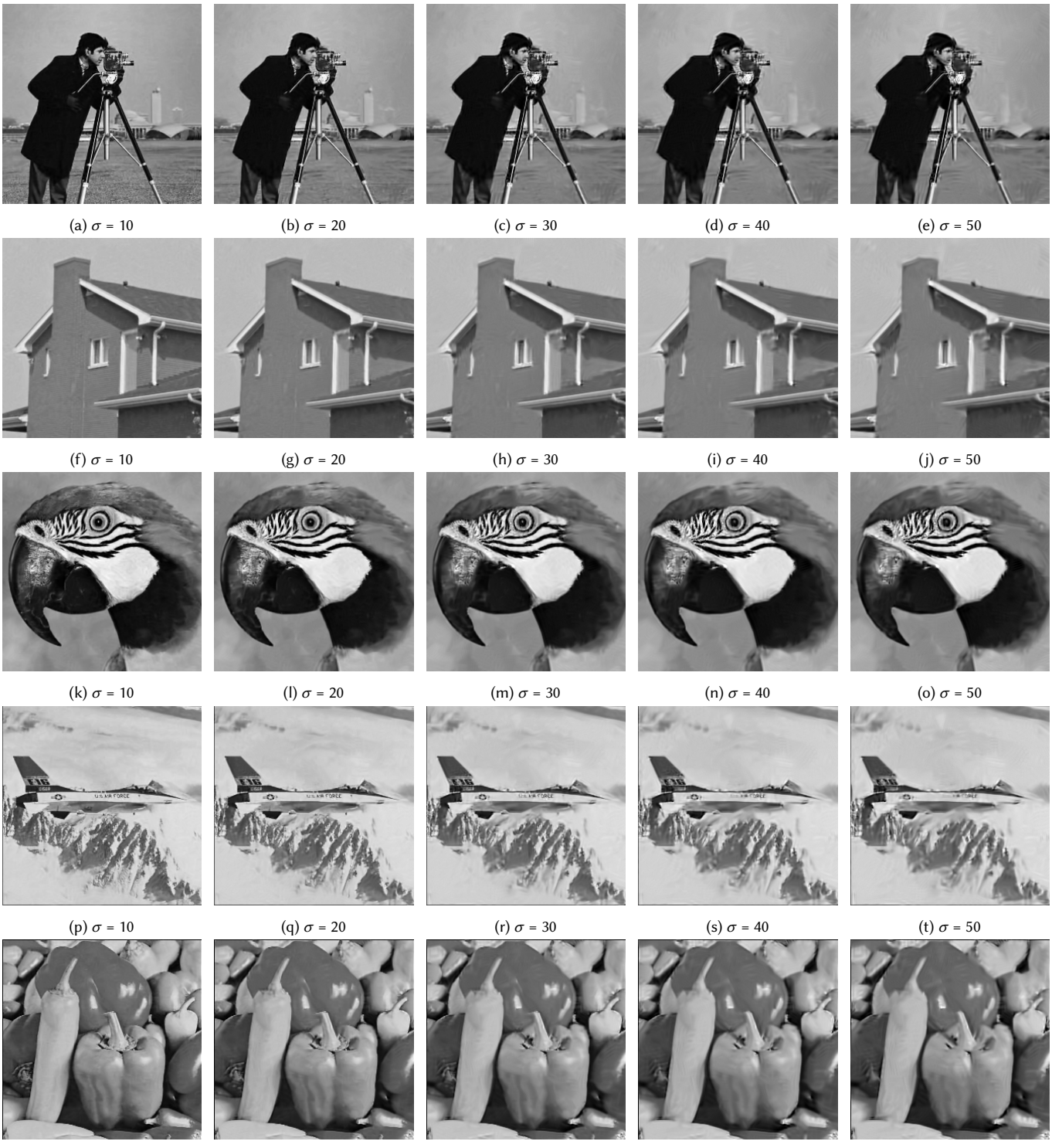
(f) σ = 10 (g) σ = 20 (h) σ = 30 (i) σ = 40 (j) σ = 50

(k) σ = 10 (l) σ = 20 (m) σ = 30 (n) σ = 40 (o) σ = 50

(p) σ = 10 (q) σ = 20 (r) σ = 30 (s) σ = 40 (t) σ = 50

(u) σ = 10 (v) σ = 20 (w) σ = 30 (x) σ = 40 (y) σ = 50

Fig. 6. Total Variation Denoised Images



(a) σ = 10 (b) σ = 20 (c) σ = 30 (d) σ = 40 (e) σ = 50

(f) σ = 10 (g) σ = 20 (h) σ = 30 (i) σ = 40 (j) σ = 50

(k) σ = 10 (l) σ = 20 (m) σ = 30 (n) σ = 40 (o) σ = 50

(p) σ = 10 (q) σ = 20 (r) σ = 30 (s) σ = 40 (t) σ = 50

(u) σ = 10 (v) σ = 20 (w) σ = 30 (x) σ = 40 (y) σ = 50

w

Fig. 7. Patch Group Based NSS Prior Denoised Image



(a) σ = 10 (b) σ = 20 (c) σ = 30 (d) σ = 40 (e) σ = 50

(f) σ = 10 (g) σ = 20 (h) σ = 30 (i) σ = 40 (j) σ = 50

(k) σ = 10 (l) σ = 20 (m) σ = 30 (n) σ = 40 (o) σ = 50

(p) σ = 10 (q) σ = 20 (r) σ = 30 (s) σ = 40 (t) σ = 50

(u) σ = 10 (v) σ = 20 (w) σ = 30 (x) σ = 40 (y) σ = 50

Fig. 8. DnCNN Denoised Images